

ORCA: Ocean Intelligence & Sea Safety

A comprehensive, plain-English guide to how space satellites, ocean physics, and artificial intelligence help Indian fishermen locate high-yield fishing zones, save diesel fuel, and navigate safely.

Prepared for: Non-Technical Users, Beginners,
Field Officers

Coverage: Arabian Sea • Bay of Bengal
• Indian EEZ

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Production)

1. What is ORCA in Simple Words?

Imagine you are a fisherman standing at the harbor in Mumbai, Kochi, or Veraval. Before you lies thousands of square kilometers of open ocean. Where should you sail your boat today?

Traditionally, captains sailed out blindly, guessing where fish might be based on intuition or luck. Often, boats sail for 10 to 18 hours, burning thousands of rupees worth of expensive diesel fuel, only to return with empty nets. Worse, sudden ocean storms, high waves, or accidental entry into restricted waters can result in tragic loss of life, confiscated vessels, or conflicts with local traditional canoes.

The Big Idea Behind ORCA

ORCA stands for **Ocean Resource & Catch Assessment**. It is an intelligent marine navigation system that acts like *"Google Maps + Live Weather Radar + Fish Finder"* for the Indian ocean. Instead of guessing, ORCA uses space satellites flying hundreds of kilometers above Earth to spot the exact underwater feeding zones where fish are naturally congregating right now.

Saves Diesel Fuel

Directs boats straight to active fish feeding grounds via calculated shortest routes, eliminating aimless offshore cruising and cutting fuel spend by up to 35%.

Protects Small Fishers

Strictly keeps commercial trawlers beyond the 12 Nautical Mile territorial reserve, preserving near-shore waters for traditional artisanal canoes.

Real-Time Sea Safety

Monitors live sea surface winds, rogue wave heights, and ocean swells so crews never sail straight into dangerous cyclonic weather.

2. How Does ORCA Know Where the Fish Are? (The Science Made Simple)

Fish don't just swim anywhere in the sea. Like cattle grazing on lush green pastures, pelagic fish (such as Tuna, Mackerel, Kingfish, and Sardines) actively search for underwater "pastures" rich in microscopic food.

SATELLITE CLUE	WHAT THE SATELLITE MEASURES	WHAT IT TELLS US ABOUT FISH
Sea Temperature (SST)	The temperature of the surface water measured by infrared satellite sensors.	Fish are cold-blooded! Commercial species prefer comfortable "thermal pockets" between 26.5°C and 29.2°C. Water that is too hot or too cold is avoided.
Chlorophyll-a (Chl-a)	The green pigment of microscopic ocean plants (phytoplankton) seen by space cameras.	More green chlorophyll = abundant tiny plant plankton = clouds of tiny shrimp & baitfish = big predatory fish arrive to feast!
Sea Height Bumps (SLA)	Satellite radar bouncing beams off the sea surface to detect centimeter-level dips and mounds.	Ocean eddies (slow-spinning underwater whirlpools) trap nutrients and forage. Depressions and ridges act like natural underwater corral fences for fish schools.
Ocean Health Index (MHI)	AI analysis of dissolved oxygen, salinity, and water chemistry.	Ensures the spot is not suffering from oxygen-starved "dead zones" (hypoxia) or industrial runoff. Healthy water = active, hungry fish.

3. Decoding the Map: What the Lines & Colors Mean

When you open the ORCA 3D Globe or 2D Map, you will see glowing dots, dotted lines, and boundaries. Here is exactly what every symbol means:

A. What the Map Lines Mean (Boundaries & Routes)

Cyan Dashed Line (---) : 12 Nautical Mile Limit (~22.2 km)

The Artisanal Fishing Reserve. Under Indian maritime law, the first 12 nautical miles from the mainland coastline are legally reserved for traditional artisanal fishermen (small unmotorized boats and wooden canoes). ORCA's AI algorithm strictly places all commercial Potential Fishing Zones (PFZs) *beyond* this 12 NM line so large motorized boats never crowd out local coastal families.

Sky Blue Dashed Line (---) : Active Navigational Route

The Transit Channel. Plotted directly from your chosen coastal departure port (e.g., Kochi Port or Mumbai Port) to your selected favorable fishing spot. It shows the safe direct channel path, straight-line distance, navigation transit time, and compass heading.

Amber Solid Line (—) : Indian EEZ Boundary (200 NM)

Exclusive Economic Zone Perimeter. Extends up to 200 nautical miles (~370 km) offshore. This is the sovereign economic zone where Indian vessels are fully authorized to harvest marine resources under national protection.

Crucial Rule: Departure Calculations are Port-to-Spot ONLY

A coastal port (like Kandla, Mundra, Mumbai, or Kochi) is a **departure base** where boats leave from. Distance calculations in ORCA are strictly calculated from a **coastal departure port to a favorable fishing spot**. You will never see route distance calculated between two coastal ports (no port-to-port routes), because fishermen sail out to catch fish, not to ferry cargo between neighboring harbors!

B. Fishing Spot Differences: What the Beacon Colors Mean

Cyan Glowing Beacon (●) : High / Favorable Potential Fishing Zone

The Jackpot Zone! (Score $\geq 68/100$, Ocean Health $\geq 42/100$). The water temperature is in the prime pelagic window (26.5–29.2°C), chlorophyll plant food is abundant ($>0.5 \text{ mg/m}^3$), and an active cold-water upwelling front is pulling baitfish to the surface. Highest probability of a rich commercial catch.

Amber Glowing Beacon (●) : Moderate / Potential Fishing Zone

Good Secondary Option. (Score $\geq 50/100$, Ocean Health $\geq 32/100$). Plankton food or water temperature is favorable, but current speeds or thermal gradients are slightly gentler. A dependable alternative if the primary cyan zone is too far from your departure port.

Slate Marker with Bronze Border (⊙) : Less Favorable Fishing Spot

Constrained / Opportunistic Zone. (Score $\geq 38/100$). Fish may be present, but certain environmental conditions are sub-optimal (e.g., water is warmer than ideal, or plankton concentration is thin). Clicking this marker opens an inspector explaining the exact *Limiting Factors* so you know why it is ranked lower.

Small Dim Gray Dot (·) : Environmental Sampling Cell (~35 km)

- **Scientific Sensor Node (NOT a Fishing Spot).** ORCA continuously monitors 500+ mathematical cells across the Indian Ocean spaced ~35 km apart. These nodes monitor water salinity, temperature, and currents. They are scientific telemetry stations, not recommended fishing destinations.

4. How to Use ORCA in 4 Easy Steps (A Captain's Checklist)

1 Pick Your Departure Port

At the bottom dock or top selector, choose where your boat is departing from (for example: *Kochi Port, Mumbai Port, Veraval Port, or Mundra Port*). You can also click directly on any coastal port marker on the map. ORCA instantly anchors your departure coordinates.

2 Locate Glowing Cyan & Amber Beacons

Look at the sea seaward of the cyan dashed 12 NM line. Spot the glowing cyan beacons (High Potential) and amber beacons (Moderate Potential). These represent the freshest, satellite-verified fish feeding hotspots.

3 Click a Beacon to Inspect Water Conditions

Clicking any beacon immediately opens the **Departure & Navigation Inspector**. Here you will see:

- **Distance in NM & km:** Measured directly from your chosen departure harbor to the spot.
- **Estimated Travel Time:** Calculated for a typical fishing trawler cruising at 10 knots.
- **Compass Bearing:** Exact nautical heading (e.g., 248° WSW) to set on your boat's compass.
- **Limiting Factors or High Drivers:** Tells you why the AI recommends this spot (e.g. strong chlorophyll, upwelling).

4 Check the Sea Safety Badge & Plot Route

Look at the **Sea Conditions Indicator** at the top right:

● **SAFE / FAVORABLE**

● **MONITORED / SWELL**

● **HAZARDOUS / HIGH SEAS**

If green, click "**Plot Route & Take Me There**". The sky-blue navigation line draws on the globe, camera zooms in, and your coordinates are ready for departure!

5. Asking ORCA AI (Your 24/7 Digital First Mate)

You don't need to be a computer expert. You can click the "**Ask ORCA**" button anytime and ask questions in plain English:

Example Questions You Can Ask:

- *"Where is the nearest high-potential tuna spot from Mumbai?"*
- *"Is it safe to sail from Kochi Port right now?"*
- *"Why is Zone 02 marked as less favorable?"*
- *"How many nautical miles to the closest fishing zone from Veraval?"*

What ORCA AI Will Answer:

ORCA will reply with the exact latitude/longitude, travel time, ocean health, and why the fish are there. It can automatically focus the 3D globe camera right onto the target beacon!

6. Beginner's Maritime Glossary

TERM	SIMPLE EXPLANATION
Nautical Mile (NM)	The standard measurement of distance at sea. 1 Nautical Mile = 1.852 kilometers . When a captain says 12 NM, that equals about 22.2 km.
Knot (kt)	The speed of a boat in nautical miles per hour. 10 knots = 18.5 km/h. Typical fishing boats travel at 8 to 12 knots.
PFZ (Potential Fishing Zone)	A specific patch of the sea identified by satellite sensors where biological and physical conditions create the highest likelihood of fish aggregations.
Upwelling Front	A natural oceanic elevator where deep, cold, nutrient-rich water pushes up toward the surface, fueling the entire marine food chain.
EEZ (Exclusive Economic Zone)	The area of coastal water up to 200 nautical miles from a country's coastline, within which the country has sole rights to natural resources.
12 NM Territorial Baseline	The sovereign waters closest to shore, reserved in India for traditional artisanal canoes to prevent industrial trawler overfishing.
SLA (Sea Level Anomaly)	Variations in ocean surface height measured in centimeters by satellite radar, revealing invisible circular ocean currents and eddies.

7. Frequently Asked Questions (FAQ)

Q: Why aren't there any recommended fishing spots right next to the beach?

A: This is an intentional feature! Indian maritime law protects the livelihood of small traditional artisanal fishermen who fish near shore in non-motorized canoes. ORCA's smart AI strictly filters commercial PFZ zones to sit outside the 12 Nautical Mile (~22.2 km) artisanal reserve.

Q: Why does clicking a coastal port show "Active Departure Base" instead of drawing a route?

A: A port is a harbor on land where your voyage starts, not a fishing destination! When you click a port (like Mumbai, Kochi, or Salaya), ORCA sets that port as your home departure base. To see a route, click any favorable fishing beacon (cyan or amber), and ORCA will instantly draw the passage from your chosen port directly to that fishing spot.

Q: What should I do if the Sea Safety badge turns Red or Amber?

A: Click on the Sea Safety badge! It will open a detailed popover explaining the exact wind speed, swell period, and wave heights. If winds exceed 25 knots or waves exceed 2.5 meters, small craft advisories are active, and vessels should avoid offshore transit.

Q: Can I use both the 3D Globe and the 2D Flat Map?

A: Yes! Click the "2D MAP / 3D GLOBE" button in the top navigation header anytime. The 2D map is great for rapid tactical panning and low-bandwidth connections, while the 3D globe provides high-precision ocean curvature and bathymetry perspective.

